

$$y(0) = C_1 + C_2$$

$$\frac{dy}{dt} = C_1 e^t - C_2 e^{-t}$$

$$\frac{d^2 y}{dt^2} = C_1 e^t + C_2 e^{-t}$$

$$\Rightarrow \text{Put in } \frac{d^2 y}{dt^2} + m \frac{dy}{dt} + n = 0$$

$$\Rightarrow C_1 e^t + C_2 e^{-t} + m(C_1 e^t - C_2 e^{-t}) + n = 0$$

$$\Rightarrow m = -1, n = 0 \quad \Rightarrow m + n = -1$$